ECE 590
Neural-Network-Based Large Language Models

Tuesdays and Thursdays 1:25-2:40
Fitzpatrick Schiciano A 1464

Housekeeping Items

Assessment:

- Homework: 40%
- Exam 1 (covering first half of semester): 20%
- Final Project (proposal, report, presentation): 40%

Classroom participation will enhance grades at the boundaries, *e.g.*, B+ → A

Teaching Assistants:

- Eric Bai
- Jiashuo Fan

Office Hours:

- (Eric) Tuesdays 3-5pm, Location: Wilkinson 320
- (Jiashuo) Fridays 2–4pm, Location: TBD

Class Roadmap - 1/4

- Over about 7 years, the Transformer architecture has gone from being invented to entirely transforming the field of natural language processing, as well as image processing and molecular analysis, and other applications

- Enormous real-world impact already (*e.g.*, ChatGPT), and likely to expand

- Field moving so quickly, with the domain of applications expanding rapidly

- In near future, Transformers for language and images may be like a *utility*, provided by several large players, like power-generation utility companies (electricity)

Class Roadmap – 2/4

- The number of people working on the fundamentals of these models will likely diminish
- Likely more activity at the application layer, at a higher level of abstraction
- Fast realignment in the type of skills valued at large tech companies
- Why study Transformer, given that fewer people are likely to work at that level in the future?

Class Roadmap – 3/4

- There will likely still be demand for skills with Transformers for targeted applications, for which the aforementioned utility model may be inappropriate

- Key for this class:

  - Transformer is an ideal training platform for almost all elements of deep neural networks and machine learning

  - For this reason, studying LLMs will provide knowledge for applications away from language, and even for generalizations of Transformers

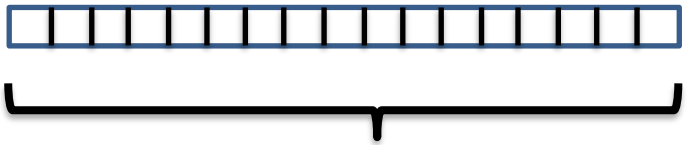
  - Will position the student for what is next

Class Roadmap – 4/4

- In this class we will use the Transformer and LLMs as a motivation to learn about many elements of machine learning that are more generally of importance
  - Few-shot learning and meta-learning
  - Reproducing kernel Hilbert spaces (RKHS) and methods for their acceleration
  - Reinforcement learning
  - Aspects of Bayesian learning
  - Focus on fundamentals, and "why?" *Why* particular design choices and methods for the Transformer? – these often generalize

Linear Models & Normalization

Learning a Linear Model Based on Labeled Data

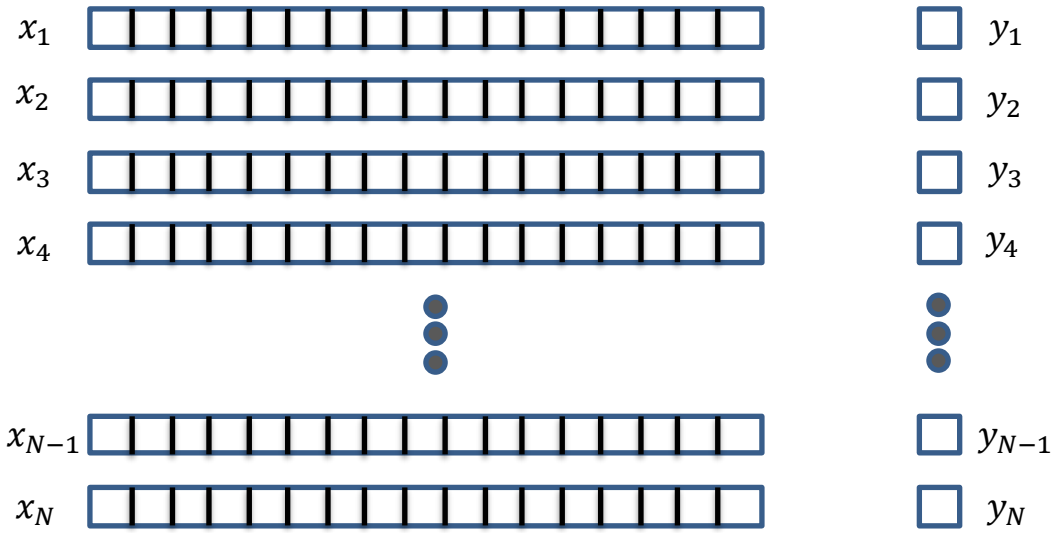


x, data/features for a subject

y, associated label 0/1

End goal: *predict* y from x

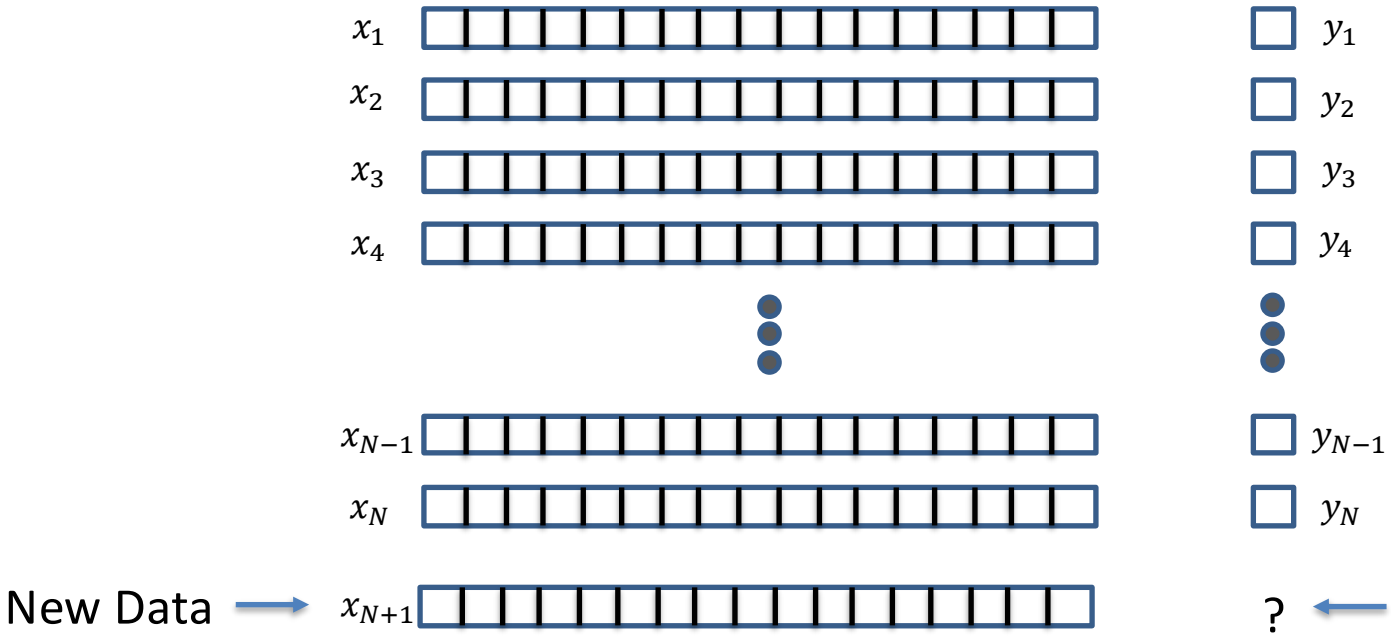
Training Set (Historical Data)



Start by limiting y to a binary outcome:

- False/True
- 0/1

Making Predictions

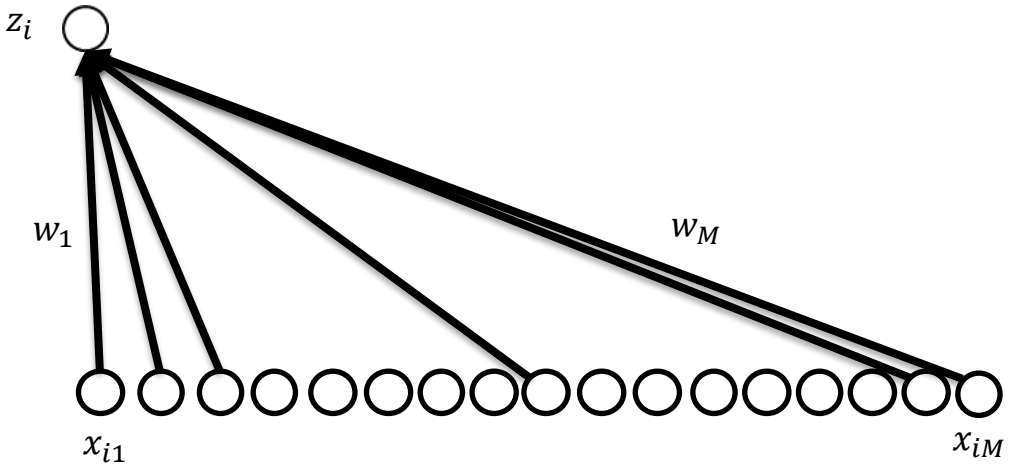


Start by limiting y to a binary outcome:

- False/True
- 0/1

Want to learn how to *predict* outcome

Linear Predictive Model



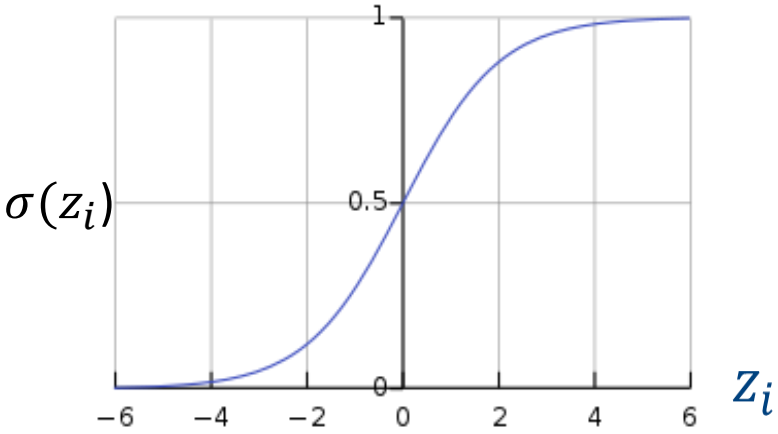
$$z_i = (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM})$$

Convert to a Probability

$$z_i = (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM}) + b$$

$$p(y_i = 1|x_i) = \sigma(z_i)$$

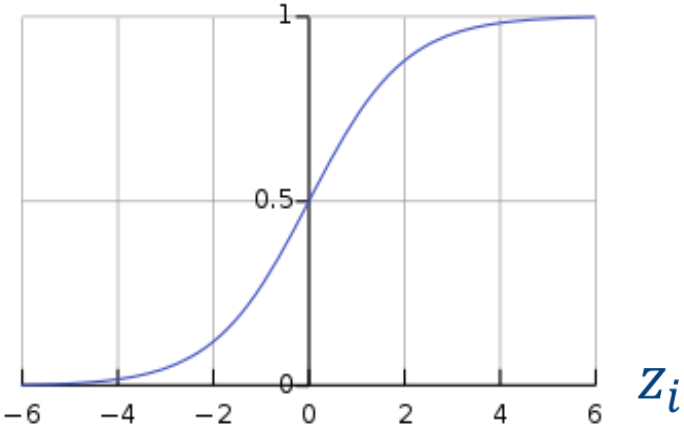
Extra Constant



Convert to a Probability

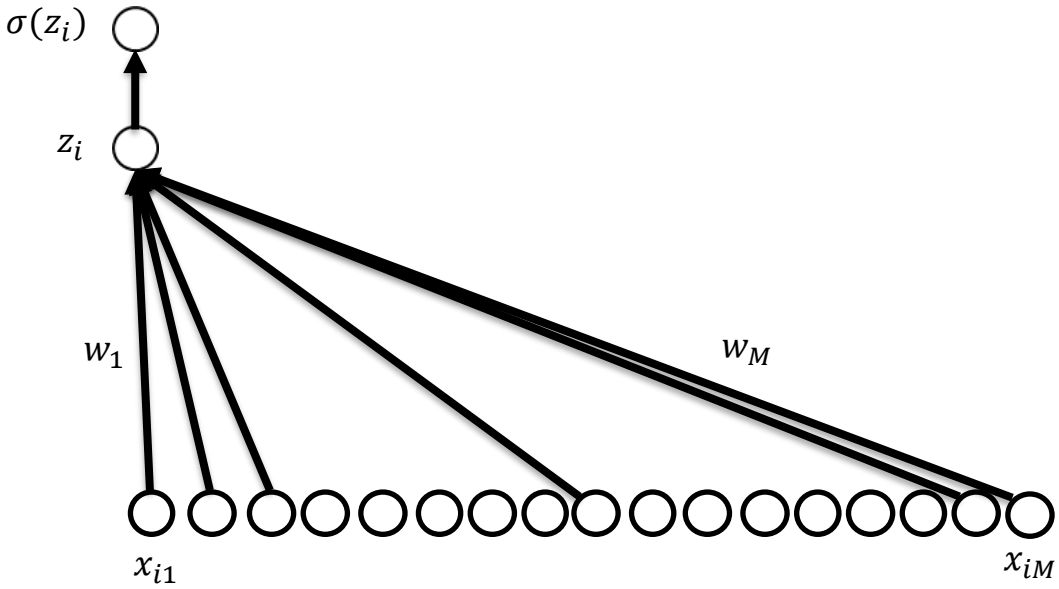
$$z_i = (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM}) + b$$

$$p(y_i = 1|x_i) = \sigma(z_i) = \frac{\exp(z_i)}{1+\exp(z_i)}$$

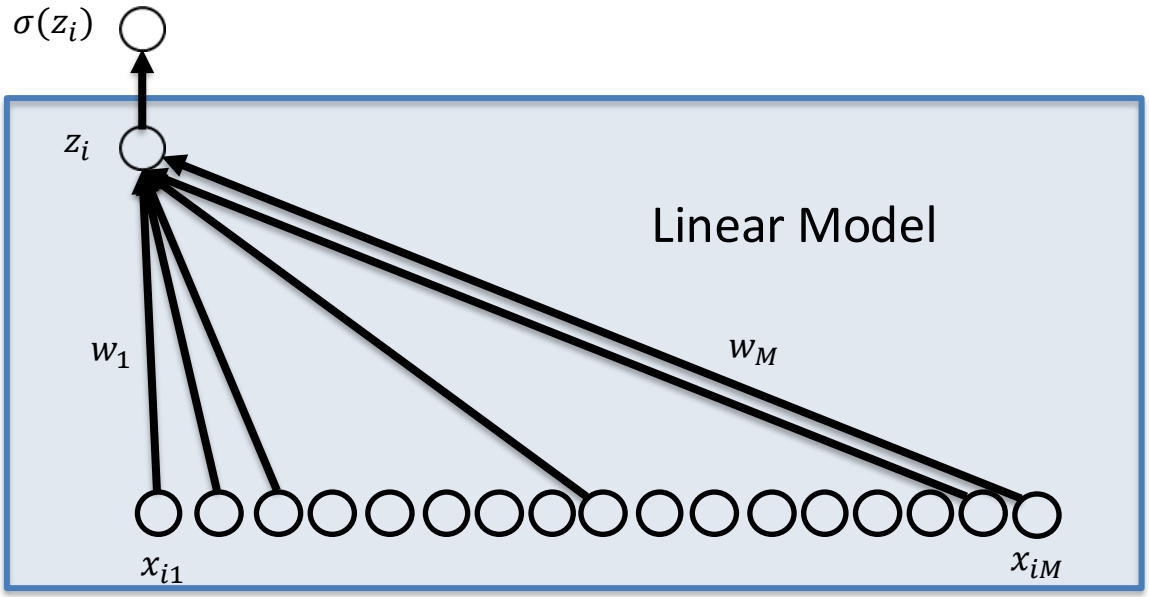


- Large and positive z_i indicates that event $y_i = 1$ is likely
- Large and negative z_i indicates that event $y_i = 0$ is likely

Logistic Regression

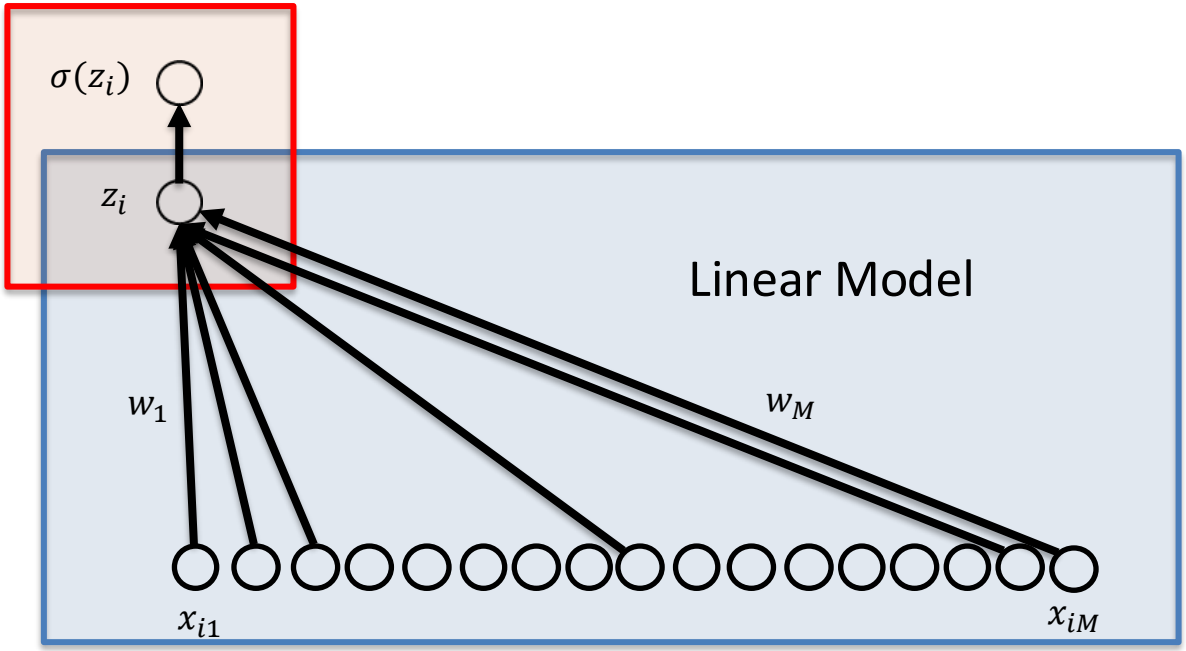


Logistic Regression



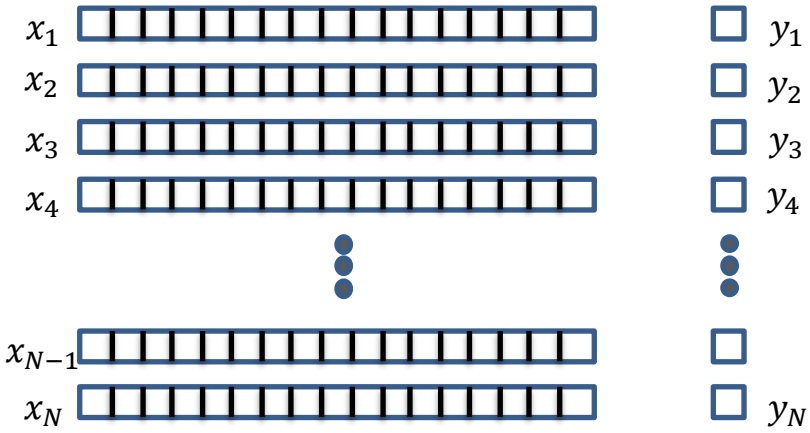
Logistic Regression

Convert to Probability (Activation)

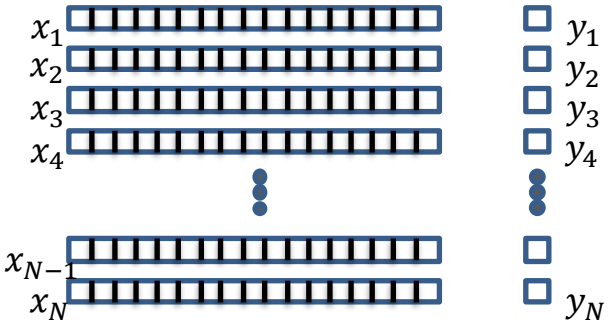


Building the Training Set

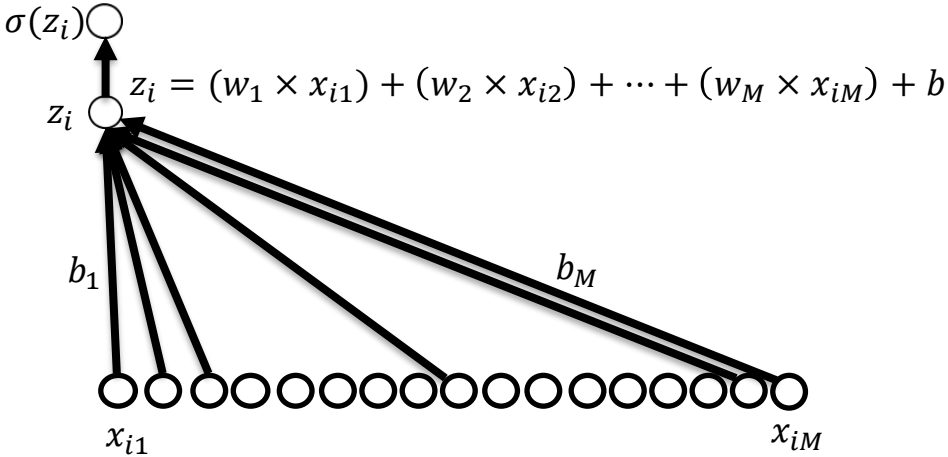
- Need to learn the parameters
- Requires *training data*
- Record data of covariates x_i and corresponding outcomes y_i



Learning Model Parameters



Training Set

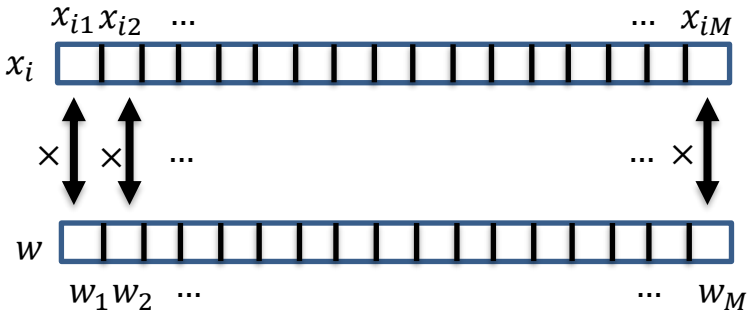


Logistic Regression Model (or "Network")

Learned Parameters $(b, w_1, \ldots w_N)$

Interpretation of Logistic Regression

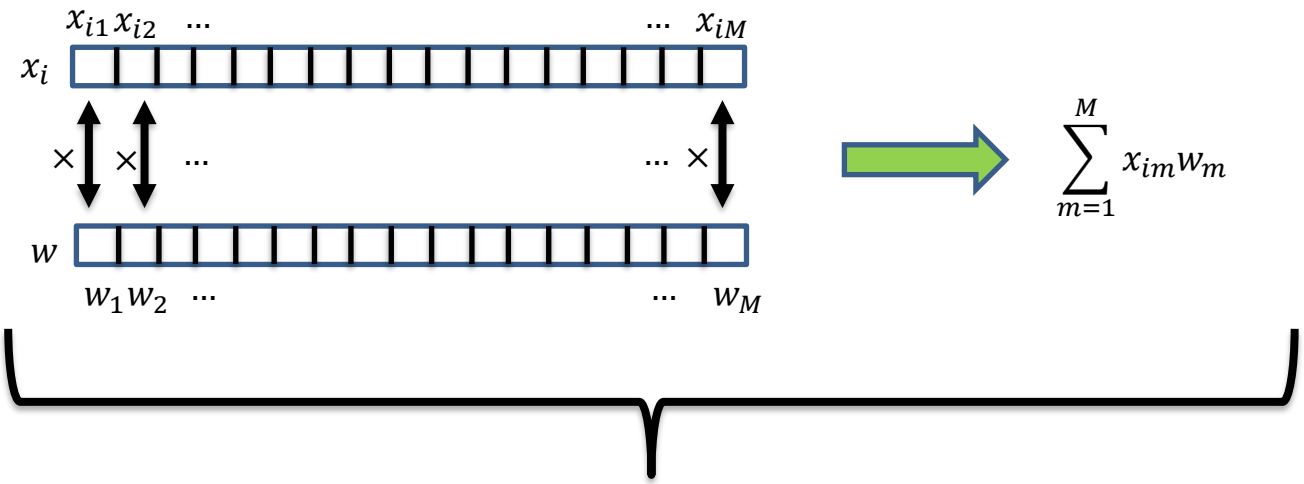
$$z_i = (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM})$$



$$\sum_{m=1}^{M} x_{im} w_m$$

Interpretation of Logistic Regression

$$z_i = (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM})$$



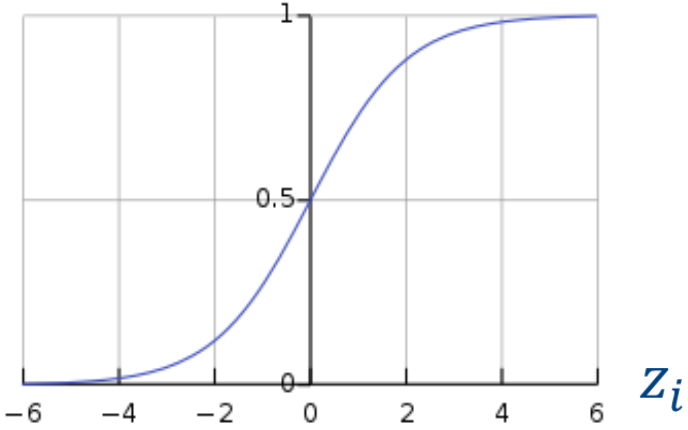
Compact Notation: $x_i \cdot w$ or $w^T x_i$

Interpretation of Logistic Regression

$$z_i = b + (w_1 \times x_{i1}) + (w_2 \times x_{i2}) + \cdots + (w_M \times x_{iM})$$

$$= b + x_i \cdot w = b + w^T x_i$$

$$p(y_i = 1|x_i) = \sigma(z_i)$$



Interpretation of Logistic Regression

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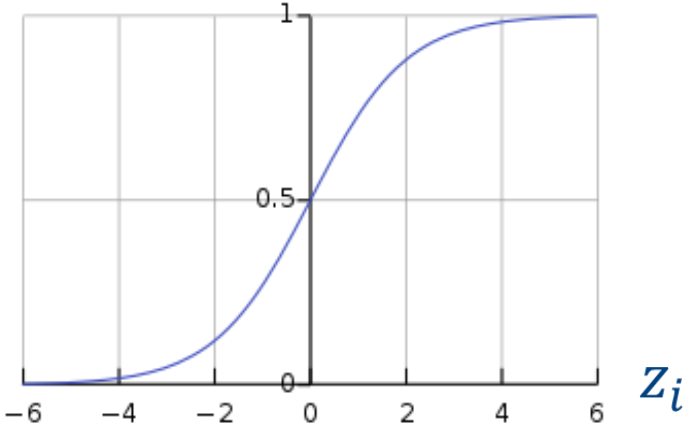


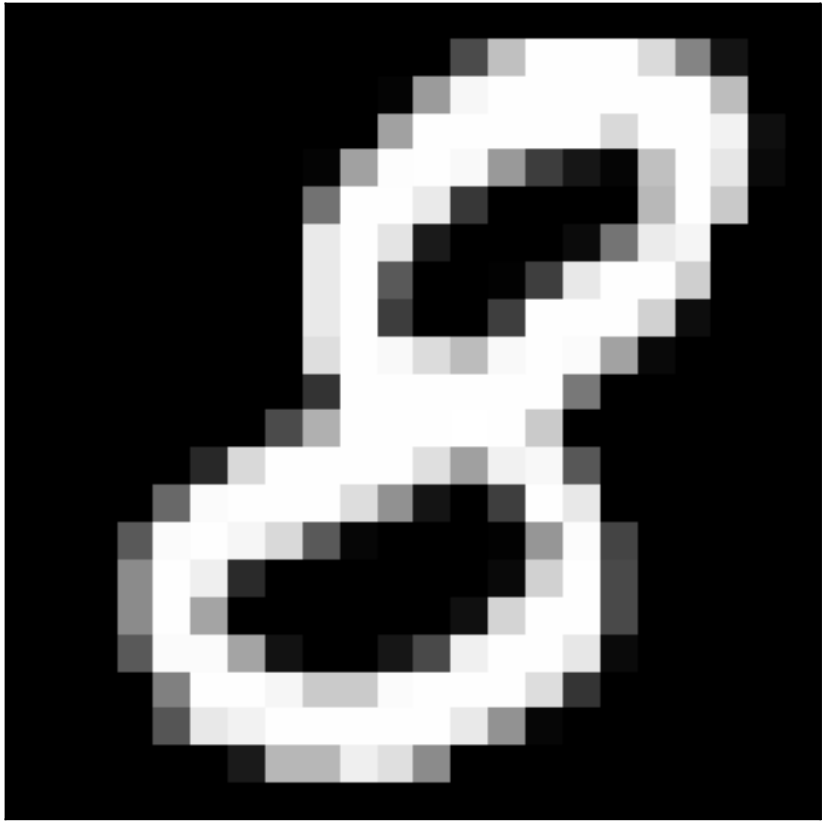
- May think of vector w as a template or filter (will visualize to make clear)
- If x_i is aligned/matched with w, then $x_i \cdot w$ will be large
- The parameter b is a bias to correct for class prevalence

RECOGNIZING NUMERICAL DIGITS

The MNIST Dataset

- The Modified National Institute of Standards and Technology (MNIST) contains pictures of handwritten digits (0,1,2,…)
- Want to be able to tell what digit each image is (*e.g.*, optical character recognition)

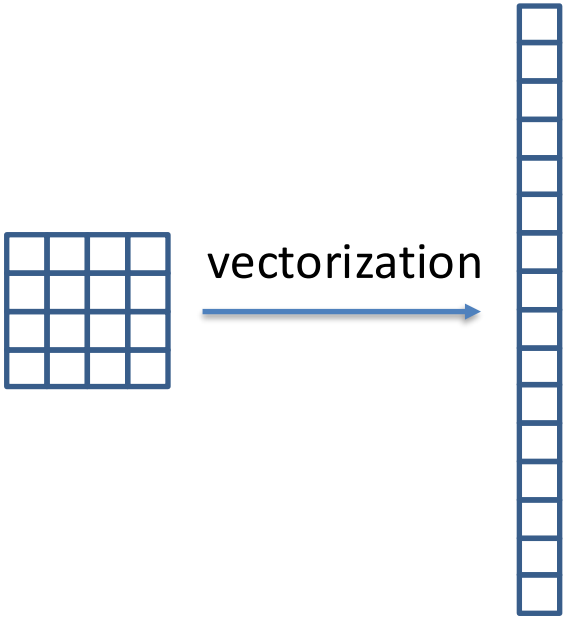
Images are Encoded as Numbers



0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	118	200	223	155	155	23	0	0	0	0	0
0	0	0	0	0	0	0	0	6	124	214	253	253	253	253	254	229	213	67	0	0	0
0	0	0	0	0	0	0	43	198	253	254	253	247	175	175	236	253	253	108	0	0	0
0	0	0	0	0	0	9	212	253	253	224	58	19	0	0	16	139	247	247	71	0	0
0	0	0	0	0	0	58	254	254	186	14	0	0	0	0	0	0	139	254	118	0	0
0	0	0	0	0	0	58	253	183	8	0	0	0	0	0	0	36	222	253	118	0	0
0	0	0	0	0	0	54	250	128	2	0	0	0	6	71	192	237	253	247	71	0	0
0	0	0	0	0	0	0	213	253	50	0	18	123	198	253	254	253	247	85	0	0	0
0	0	0	0	0	0	0	69	241	227	136	200	253	253	253	254	192	34	0	0	0	0
0	0	0	0	0	0	0	0	178	254	256	254	254	254	149	59	0	0	0	0	0	0
0	0	0	0	0	0	0	101	253	253	254	253	253	253	42	0	0	0	0	0	0	0
0	0	0	0	0	8	138	247	253	243	159	196	243	253	199	0	0	0	0	0	0	0
0	0	0	0	12	183	253	253	253	50	0	0	49	253	214	0	0	0	0	0	0	0
0	0	0	0	180	254	253	215	50	2	0	0	71	253	214	0	0	0	0	0	0	0
0	0	0	0	234	217	97	10	0	0	0	23	207	254	215	0	0	0	0	0	0	0
0	0	0	174	253	156	0	0	0	0	45	215	253	253	95	0	0	0	0	0	0	0
0	0	9	210	253	163	5	19	49	130	244	253	251	137	4	0	0	0	0	0	0	0
0	0	13	229	253	254	192	253	253	253	254	253	137	0	0	0	0	0	0	0	0	0
0	0	0	160	253	254	253	253	253	253	239	132	4	0	0	0	0	0	0	0	0	0
0	0	0	18	112	194	254	254	254	163	59	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

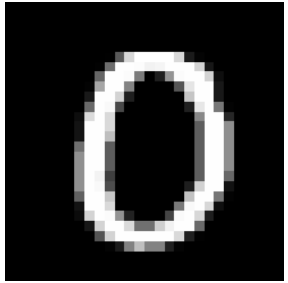
Vectorization

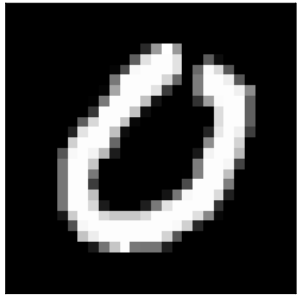
- We will start talking about deep learning *without* using the structure of the image
- We will return to this in a future lecture
- To convert an image into an unstructured set of numbers, we *vectorize it*

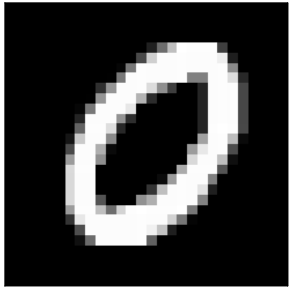


Start With The Binary Case

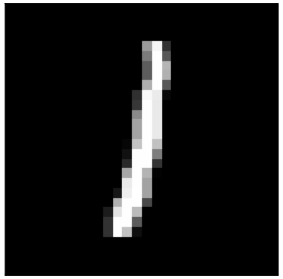
Zeros

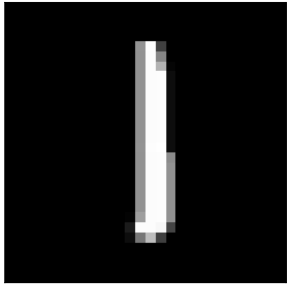


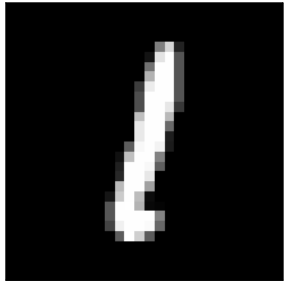




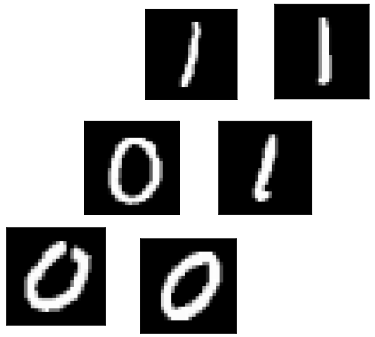
Ones



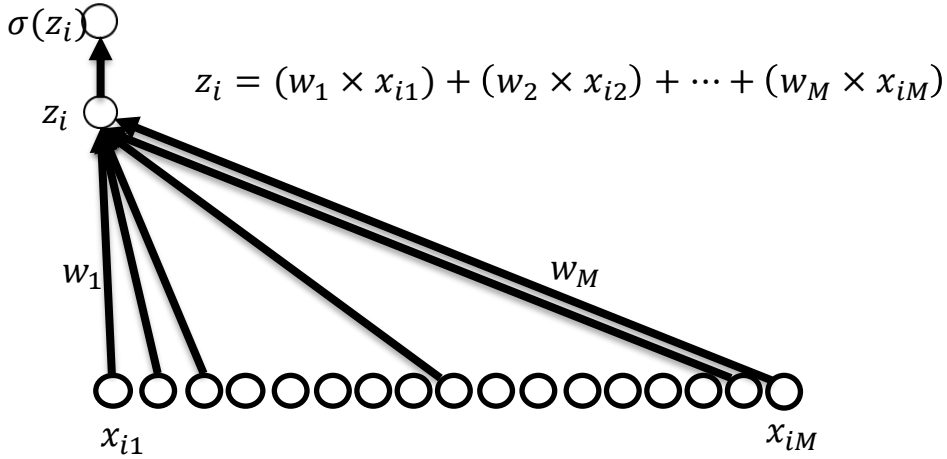




Learning on MNIST



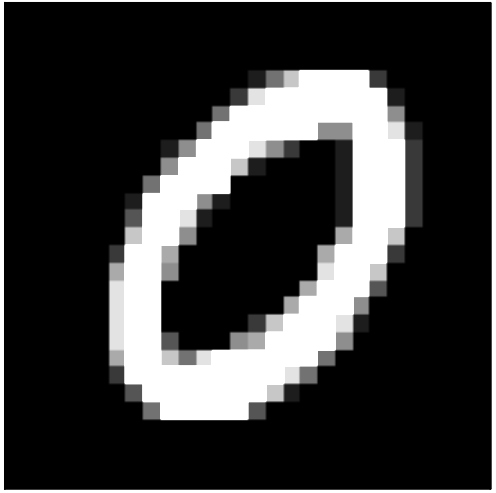
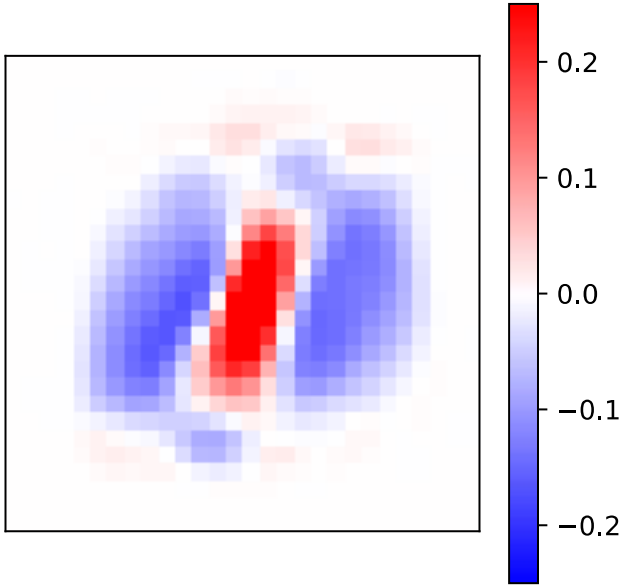
Vectorization



Logistic Regression Model (or "Network")

Learned Parameters $(b, w_1, \ldots w_M)$

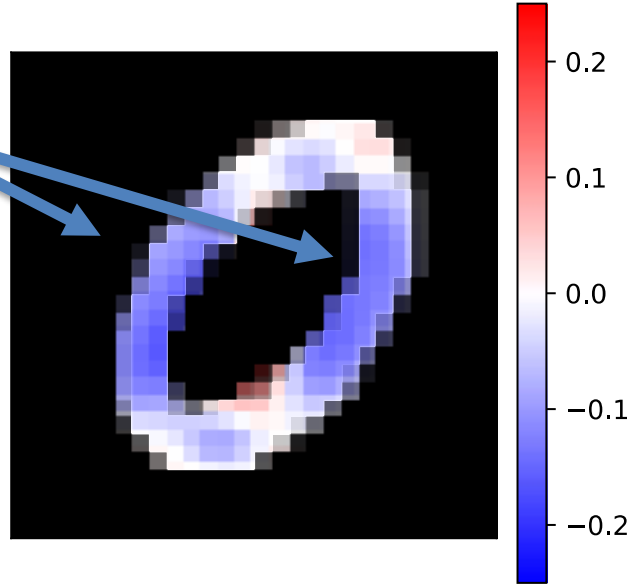
Zooming in on 0/1

$\sigma($  $\times$ 

$)$

Zooming in on 0/1

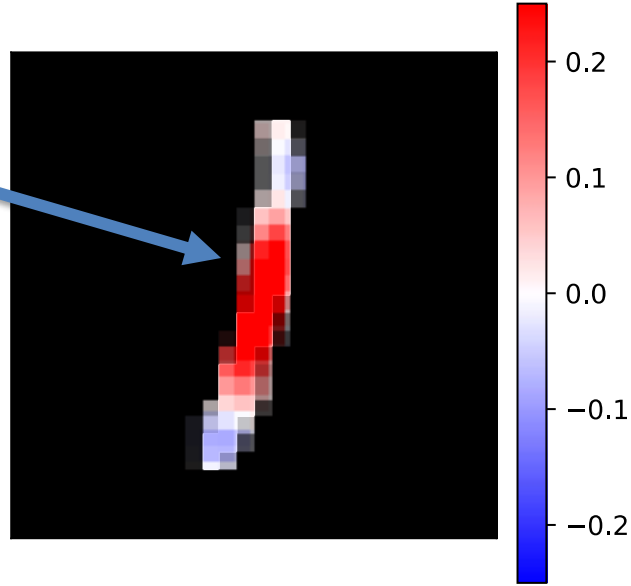
Negative Sections

$\sigma($ 

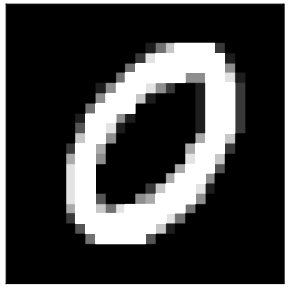
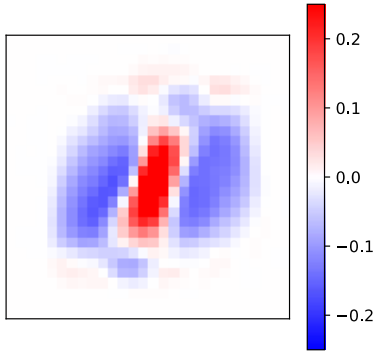
$) = 0.006$

Zooming in on 0/1

Positive Section

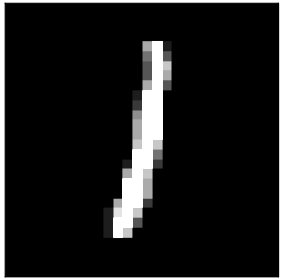
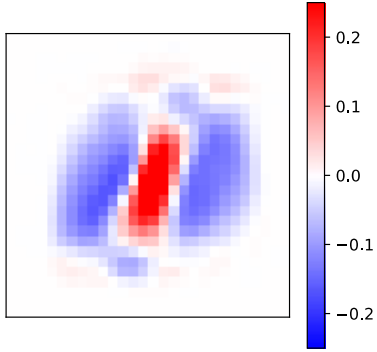
$\sigma($  $) = .991$

Learned Weights for 0/1

$\sigma($  $\times$ 

$) = .006$

We think that this is a “zero” (0.6% chance it is a “one”)

$\sigma($  $\times$ 

$) = .991$

We think that this is a “one” (99.1% chance it is a “one”)

SLOWING DOWN: WHAT'S A LINEAR MODEL?

- The concept of "linear" will have multiple different notions:

  1. Project the data x onto a line, defined by vector w

  2. A linear model of the outcome along such line

  3. Equal predicted outcomes, for points along lines perpendicular to w

- Next time we will walk through this stepwise